Applied Unsupervised Learning in Python

Today's goal • 3/3 items

Module 1
Basic Unsupervised Learning Methods

Module 2
Clustering

Module 3
Unsupervised Methods for Text Analysis

Module 4
Applications and Variants of Unsupervised Learning

Lesson 1: Unsupervised and Supervised Learning
- Applying Unsupervised Learning to Supervised Learning Tasks
  Video • 19 min
- Time to Practice: Applying Unsupervised Learning to Supervised Learning Tasks
  Graded Assignment • Grade: 100%

Lesson 2: Handling Missing Data
- Imputation of Missing Data
  Video • 14 min
- Imputation with Scikit-Learn
  Video • 19 min
- Time to Practice: Imputation of Missing Data
  Graded Assignment • Grade: 100%

Lesson 3: Semi-Supervised Learning
- A Brief Introduction to Semi-Supervised Learning
  Video • 19 min
- Label propagation with scikit-learn
  Video • 15 min
- Time to Practice: Semi-Supervised Learning
  Graded Assignment • 15 min

Lesson 4: Self-Supervised Learning
- A Brief Introduction to Self-Supervised Learning
  Video • 17 min
- Time to Practice: Self-Supervised Learning
  Graded Assignment • 15 min

Lesson 5: Module 4 Wrap-Up
- **Introduction to Module 4 Assignment: Applying Methods and Techniques for Data Imputation and Semi-Supervised Learning**
  Reading • 5 min
- Create & Submit Module 4 Assignment
  Programming Assignment • Grade: 100%
- Assignment 4 Followup
  Dialogue • 15 min
- Module 4 Optional Readings & Resources
  Reading • 5 min

Introduction to Module 4 Assignment: Applying Methods and Techniques for Data Imputation and Semi-Supervised Learning

0:35 / 0:43 2x

For the Module 4 programming assignment, you will discover how supervised and unsupervised learning can be used in conjunction with each other for important data science tasks. You will begin with data imputation and move on to an example of conducting semi-supervised learning with label propagation.

1. **First,** you are going to test out and compare several different imputation strategies to deal with missing values of the *features* in a dataset.
2. **Second**, you will do label propagation on a partially-labelled dataset of mushrooms to determine their edibility and compute the accuracy of the label propagation prediction (i.e. the accuracy of mushrooms being labeled as edible/unedible).

Go to next item ✓ Completed

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